

Answer all questions in the spaces provided.

Do not write
outside the
box

- 1 The equation of a curve is given by

$$y = 3e^x$$

Find an expression for $\frac{dy}{dx}$

Circle your answer.

AQA A LEVEL MATHS

PAPER 1 2025

FULL MARK SCHEME

Which **one** of the following can be used to describe this sequence?

Circle your answer.

[1 mark]

Convergent

Decreasing

Increasing

Periodic

the sequence is a proaching 
but its signs alternate so it is
NOT decreasing.

3

Express

$$\log_3 2x - \log_3 x$$

as a single logarithm.

Circle your answer.

[1 mark]

$\log_3 \frac{1}{2}$

$\log_3 2$

$\log_3 x$

$\log_3 2x^2$

Turn over for the next question

$$\log_3 \left(\frac{2x}{x} \right) = \log_3 (2)$$

- 4 The first three terms, in ascending powers of x , of the binomial expansion of $(1 - 8x)^{\frac{1}{2}}$ are

$$1 + nx - 8x^2$$

where n is a constant.

- 4 (a) State the range of values of x for which the expansion is valid.

Circle your answer.

[1 mark]

$$|x| > -8 \quad |x| > -\frac{1}{8} \quad |x| < \frac{1}{8} \quad |x| < 8$$

- 4 (b) State the value of the constant n

Circle your answer.

[1 mark]

$$-16$$

$$-4$$

$$\frac{1}{2}$$

$$4$$

$$(p + qx)^n \text{ valid for } |x| < \left| \frac{p}{q} \right|$$

$$|x| < \left| \frac{1}{-8} \right|$$

$$|x| < \frac{1}{8}$$

5

Find $\int (4x - 3 + x^{-\frac{1}{2}}) dx$

[3 marks]

$$\int 4x - 3 + x^{-\frac{1}{2}} dx$$

$$= 2x^2 - 3x + 2x^{\frac{1}{2}} + C$$

6

An arithmetic series has first term 3 and common difference -0.5

Find the sum of the first 250 terms.

[2 marks]

$$a = 3 \quad d = -0.5 \quad n = 250$$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$= \frac{250}{2} [6 + (249 \times -0.5)]$$

$$= -14812.5$$

7

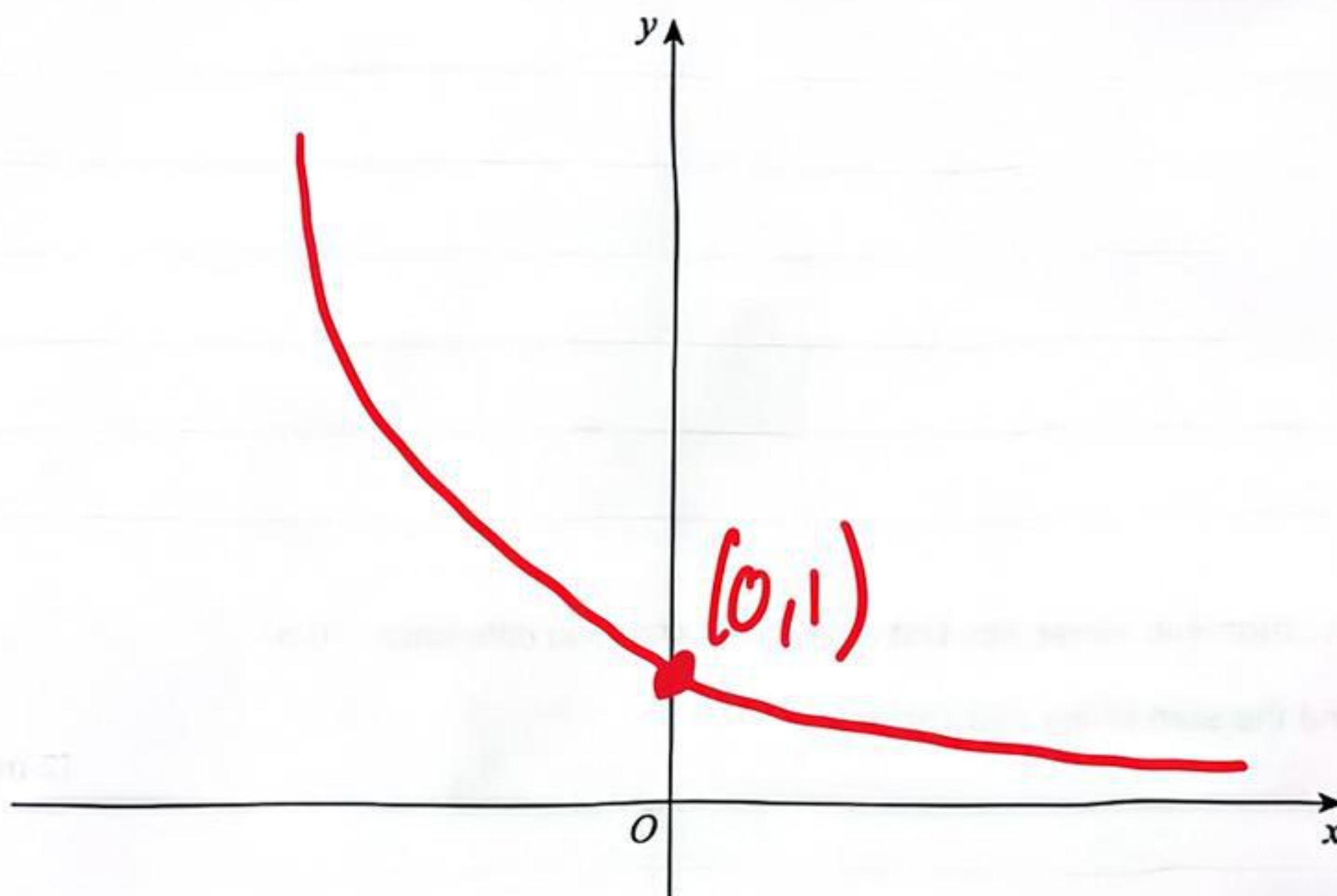
It is given that $0 < a < 1$

Sketch the graph with equation

$$y = a^x$$

on the axes below.

[2 marks]



if $0 < a < 1$ a is fractional.
 Therefore it can be considered
 as a^{-x} for $a > 1$.
 if $f(x) = a^x$, $a^{-x} = f(-x)$
 $f(-x)$ is a reflection in y axis

- 9 (a) A geometric series, S , has **second** term 60

The common ratio of S is 0.2

Find the exact value of the sum of the first five terms of S

[4 marks]

$$S_n = \frac{a(1-r^n)}{1-r}$$

$$r = 0.2 \quad n = 5 \quad \text{if term 2} = 60$$

$$\text{It means term 1} \times 0.2 = 60$$

$$\text{So } a = 60 \div 0.2 = 300$$

$$S_5 = \frac{300(1-0.2^5)}{1-0.2} = 374.88$$

- 9 (b) A different geometric series, T , has **second** term 60 and positive common ratio r

The sum to infinity of T is T_∞

$$a = \frac{60}{r}$$

$$S_\infty = \frac{a}{1-r}$$

- 9 (b) (i) Show that

$$T_\infty = \frac{60}{r-r^2}$$

[2 marks]

$$\begin{aligned} \frac{\left(\frac{60}{r}\right)}{1-r} &= \frac{60}{r} \div (1-r) \\ &= \frac{60}{r} \times \frac{1}{1-r} = \frac{60}{r-r^2} \end{aligned}$$

9 (b) (ii) Find the maximum value of $r - r^2$

[2 marks]

We are told r is positive
and the sequence converges.
Since it has a sum to infinity.

Therefore $0 < r < 1$ Complete Square

$$r - r^2 = -\left(r - \frac{1}{2}\right)^2 + \frac{1}{4}$$

max val = $\frac{1}{4}$ when $r = \frac{1}{2}$

Could differentiate too.

9 (b) (iii) Hence find the range of possible values of T_{∞}

Fully justify your answer.

[2 marks]

$T_{\infty} = \frac{60}{r - r^2}$, this will take its

minimum value when denom = max = $\frac{1}{4}$

$$T_{\infty \text{ min}} = \frac{60}{0.25} = 240$$

$$T_{\infty} > 240$$

- 10 A researcher working for a frozen-food manufacturer uses the formula

$$\theta = 21 - Ae^{-kt}$$

to model the temperature of a dessert once it is taken out of a freezer.

In this model:

- θ is the temperature of the dessert in $^{\circ}\text{C}$
- t is the time in hours since the dessert was removed from the freezer
- A and k are positive constants.

- 10 (a) Show how

$$\theta = 21 - Ae^{-kt}$$

can be rearranged to obtain

$$\ln(21 - \theta) = -kt + \ln A$$

[3 marks]

$$\theta = 21 - Ae^{-kt}$$

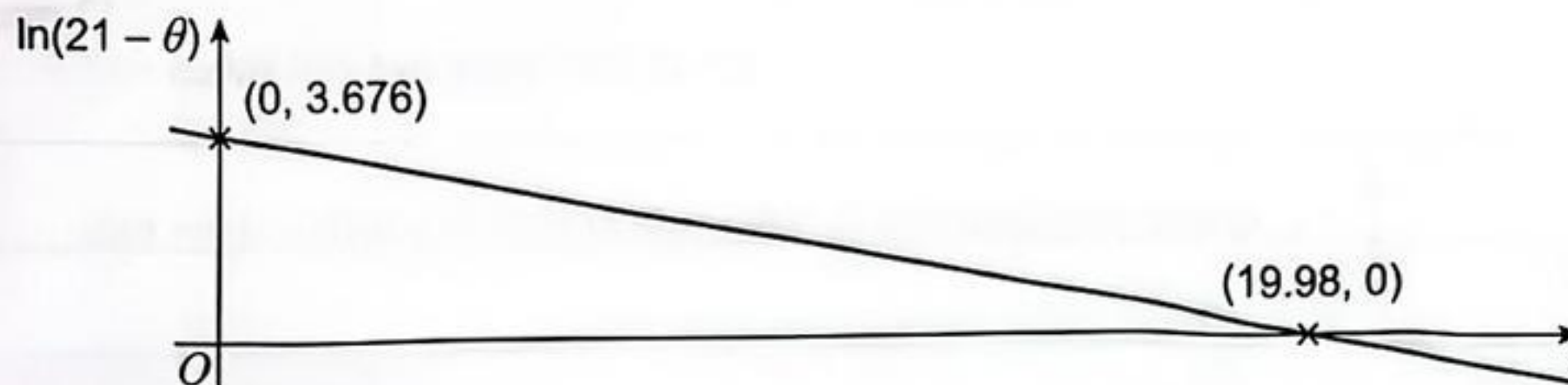
$$21 - \theta = Ae^{-kt}$$

$$\ln(21 - \theta) = \ln(Ae^{-kt})$$

$$\ln(21 - \theta) = \ln A + \ln(e^{-kt})$$

$$\ln(21 - \theta) = \ln A - kt$$

- 10 (b) The researcher uses measurements they have recorded to plot the graph of $\ln(21 - \theta)$ against t as shown in the diagram below.



- 10 (b) (i) Use the information on the graph to find the value of A

Give your answer to **three** significant figures.

[2 marks]

$$\ln(21 - \theta) = \ln A - kt$$

$$y = c + mx$$

$$3.676 = \ln A$$

$$A = e^{3.676} = 39.5$$

- 10 (b) (ii) Use the information on the graph to find the value of k

Give your answer to **three** significant figures.

[2 marks]

find gradient $\frac{3.676 - 0}{0 - 19.98}$

$$= -0.184$$

$$k = 0.184$$

10 (b) (iii) Find the temperature of the dessert when it is initially removed from the freezer.

Give your answer to three significant figures.

[2 marks]

Can use linear relationship or original model.

$$\theta = 21 - 39.5e^{-0.184t}$$

When $t=0$ $\theta = 21 - 39.5$
 $\theta = -18.5$

[2 marks]

10 (c) The dessert is ready to be eaten when its temperature reaches 4°C

Use the model to determine the time, after being removed from the freezer, for the dessert to reach this temperature.

Give your answer to the nearest 10 minutes.

[3 marks]

$$4 = 21 - 39.5e^{-0.184t}$$

$$e^{-0.184t} = \frac{17}{39.5}$$

$$-0.184t = \ln\left(\frac{17}{39.5}\right)$$

$$t = \frac{-1}{0.184} \ln\left(\frac{17}{39.5}\right) = 4.582 \text{ (3dp)}$$

4.582 hrs 0.582 hrs = 0.582×60 mins
 $= 34.92$
 4 hrs 30 mins (nearest 10)

11

The equation of a curve is

$$x^2y + 4y^3 = 8x$$

The curve has two stationary points.

11 (a)

Use implicit differentiation to show that at the stationary points $y = \frac{4}{x}$

[4 marks]

Product rule for first term

$$2xy + x^2 \frac{dy}{dx} + 12y^2 \frac{dy}{dx} = 8$$

at stationary points $\frac{dy}{dx} = 0$

$$\text{So } 2xy = 8$$

$$y = \frac{8}{2x}$$

$$y = \frac{4}{x}$$

11 (b)

Hence show that the x -coordinates of the stationary points can be written in the form $\pm\sqrt{n}$ where n is an integer to be found.

[3 marks]

Swap y in Curve equation

$$x^2\left(\frac{4}{x}\right) + 4\left(\frac{4}{x}\right)^3 = 8x$$

$$4x + \frac{256}{x^3} = 8x$$

$$4x^4 + 256 = 8x^4 \quad 4x^4 = 256$$

$$x^4 = 64 \quad x^2 = 8$$

$$x = \pm\sqrt{8}$$

12

Functions f and g are defined by

$$\begin{aligned} f(x) &= x^2 + 5 & x &\in \mathbb{R} \\ g(x) &= \sqrt{x} & x &\geq 0 \end{aligned}$$

12 (a)

Using set notation, state the range of f

[2 marks]

a quadratic has a minimum and no maximum. range will be $f(x) \geq \min$
 min value = 5 so $f(x) \geq 5$

Set notation: $\{f(x) \in \mathbb{R} : f(x) \geq 5\}$

12 (b)

Determine whether f has an inverse.

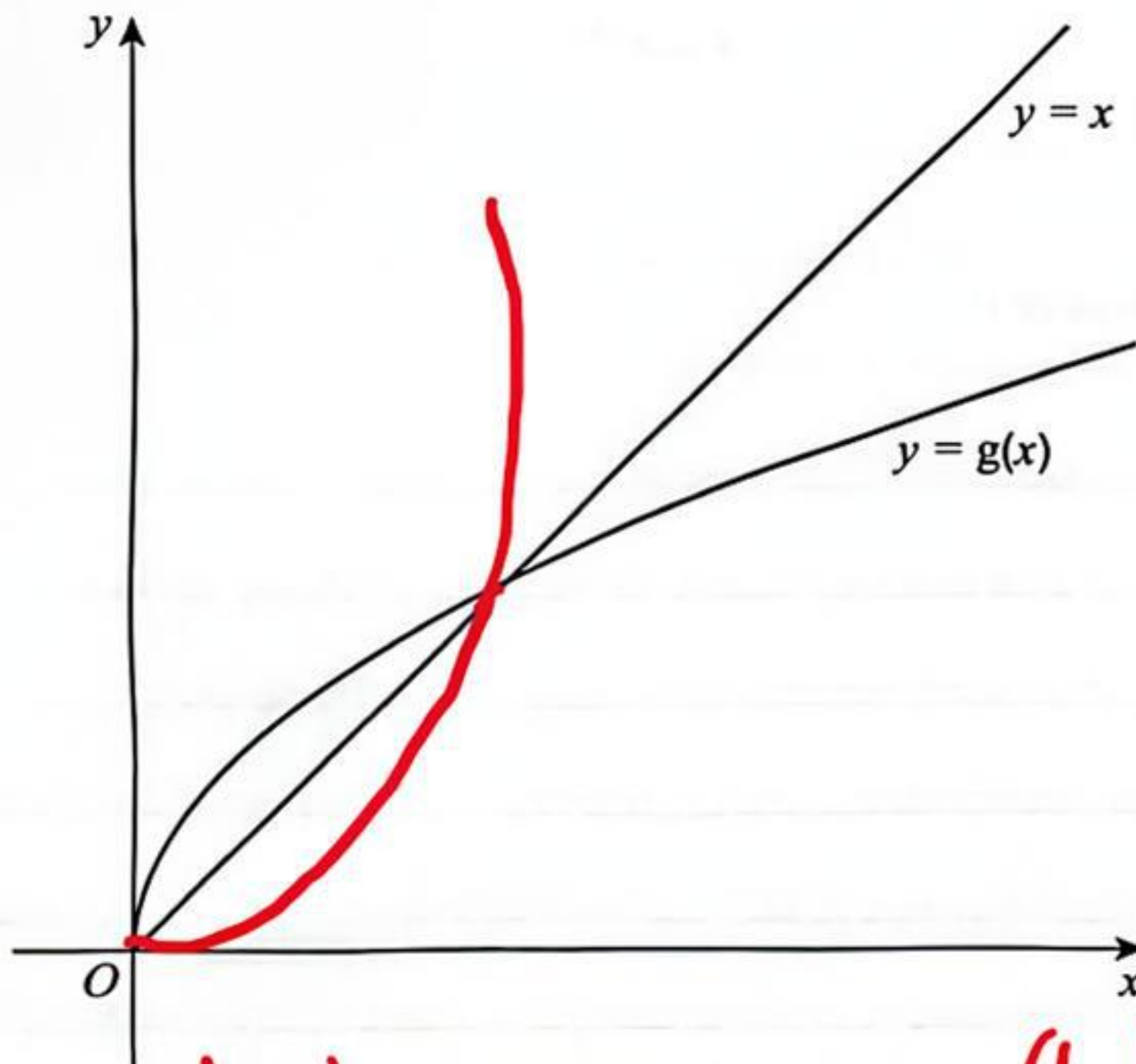
Fully justify your answer.

[2 marks]

the function is defined over all
 real numbers. As such, it is a
 many to one function. Many to one
 functions do NOT have inverses.

12 (c)

The graph of $y = g(x)$, and the line with equation $y = x$, are shown on the diagram below.



A function and its inverse are reflections in $y = x$

Sketch the graph of $y = g^{-1}(x)$ on the diagram.

[2 marks]

12 (d) The composite function gf is denoted by h

12 (d) (i) Write down an expression for $h(x)$

[1 mark]

$$f(x) = x^2 + 5 \quad g(x) = \sqrt{x}$$

$$gf(x) = \sqrt{x^2 + 5}$$

12 (d) (ii) State the range of h

[1 mark]

A square root value increases as the value you are square rooting increases.
 the min of $x^2 + 5$ is 5 so min of $\sqrt{x^2 + 5} = \sqrt{5}$
 $h(x) \geq \sqrt{5}$

13

A curve C has parametric equations

Chain rule for $\frac{dy}{dx}$

$$x = 4(4t + 1)^2$$

$$y = e^{-4t}$$

for $-\frac{1}{4} \leq t \leq 0$

$$\frac{dx}{dt} = 32(4t+1)$$

$$\frac{dy}{dt} = -4e^{-4t}$$

13 (a)

Find $\frac{dy}{dx}$ in terms of t

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

[3 marks]

$$\frac{dy}{dx} = -4e^{-4t} \times \frac{1}{32(4t+1)}$$

$$\frac{dy}{dx} = \frac{-4e^{-4t}}{32(4t+1)} = \frac{-e^{-4t}}{8(4t+1)}$$

13 (b)

Find an equation of the tangent to C at the point where $t = 0$

$$y = mx + c$$

[3 marks]

gradient of tangent = gradient of curve
at point of contact.

$$\frac{dy}{dx} = \frac{-4}{32} = \frac{-1}{8} \text{ when } t = 0$$

$$x = 4 \quad y = 1 \quad 1 = \left(\frac{-1}{8} \times 4\right) + c \quad c = \frac{3}{2}$$

$$y = \frac{-1}{8}x + \frac{3}{2}$$

13 (c)

Find a Cartesian equation for C in the form $y = f(x)$

Fully justify your answer.

[3 marks]

$$x = 4(4t+1)^2 \quad y = e^{-4t}$$

Isolate a t and Sub in

$$\frac{1}{4} \left(\sqrt{\frac{x}{4}} - 1 \right) = t$$

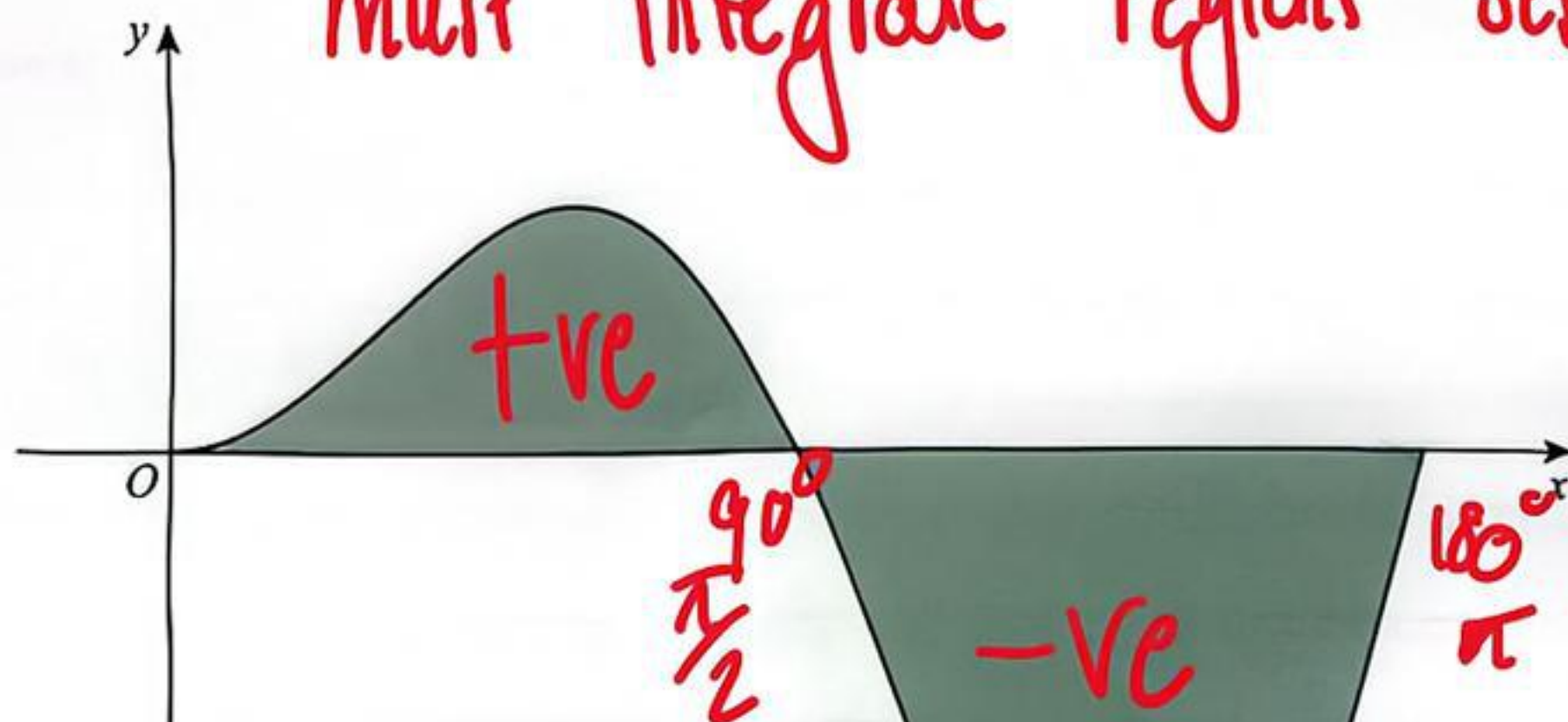
$$y = e^{-4 \left(\frac{1}{4} \left(\sqrt{\frac{x}{4}} - 1 \right) \right)}$$

$$y = e^{- \left(\sqrt{\frac{x}{4}} - 1 \right)}$$

$$y = e^{1 - \sqrt{\frac{x}{4}}}$$

Turn over for the next question

$$\text{or } y = e^{1 - \frac{\sqrt{x}}{2}}$$



$\int 4x \sin 2x dx$ by parts $\int v \frac{du}{dx} = uv - \int u \frac{dv}{dx}$

$v = 4x \quad \frac{dv}{dx} = \sin 2x \quad \frac{du}{dx} = 4 \quad u = -\frac{1}{2} \cos 2x$

Show that the shaded area enclosed by the x -axis and the curve is $k\pi$, where k is an integer to be found.

Fully justify your answer.

$$\int 4x \sin 2x = -2x \cos 2x - \int -2 \cos 2x \quad [8 \text{ marks}]$$

$$= -2x \cos 2x + \sin 2x$$

$$[-2x \cos 2x + \sin 2x]_0^{\pi/2} = -\pi \cos(\pi) + \sin(\pi) = \pi$$

$$[-2x \cos 2x + \sin 2x]_{\pi/2}^{\pi} = [-2\pi \cos(2\pi) + \sin(2\pi)] - (-\pi \cos \pi + \sin \pi) = (-2\pi + 0) - (-\pi + 0) = -\pi$$

$$\text{total area} = \pi + 3\pi = 4\pi$$

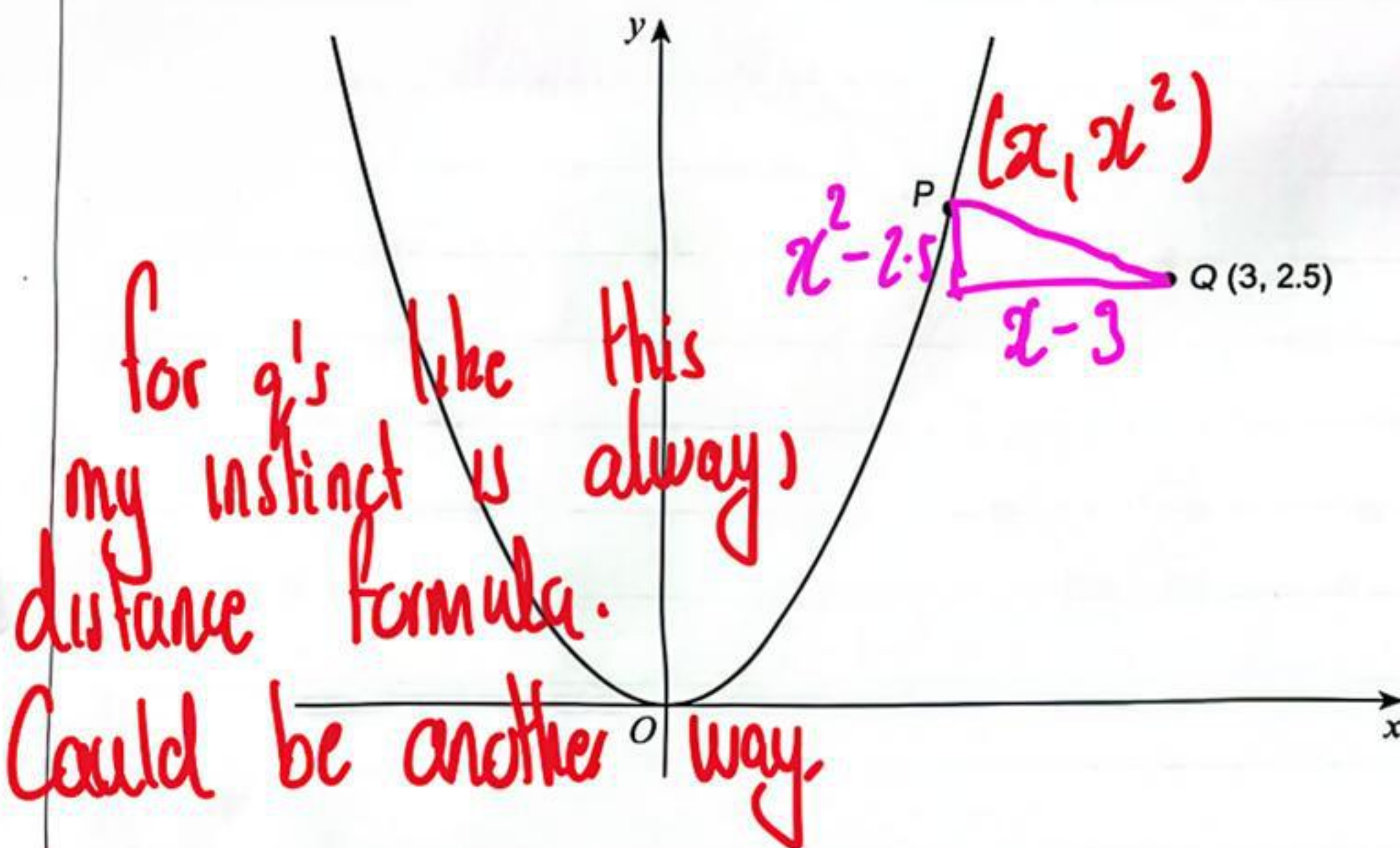
15

A curve has equation

$$y = x^2$$

The point Q has coordinates (3, 2.5)

The point P on the curve which is closest to the point Q is shown on the diagram below.



15 (a)

Show that the x-coordinate of P satisfies the equation

$$2x^3 - 4x - 3 = 0$$

$$d^2 = (x^2 - 2.5)^2 + (x - 3)^2$$

[4 marks]

$$= (x^4 - 5x^2 + 6.25) + (x^2 - 6x + 9)$$

$$= x^4 - 4x^2 - 6x + 15.25$$

If closest to Q, distance is minimum so
derivative = 0

$$4x^3 - 8x - 6 = 0$$

$$2x^3 - 4x - 3 = 0$$

15 (b)

The Newton-Raphson method is to be used to find an approximate solution to the equation

$$2x^3 - 4x - 3 = 0$$

Show that the Newton-Raphson method generates the iterative formula

$$x_{n+1} = \frac{4x_n^3 + 3}{6x_n^2 - 4}$$

[4 marks]

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f(x_n) = 2x_n^3 - 4x_n - 3 \quad f'(x_n) = 6x_n^2 - 4$$

$$x_{n+1} = x_n - \frac{2x_n^3 - 4x_n - 3}{6x_n^2 - 4}$$

Common denom

$$x_{n+1} = \frac{x_n(6x_n^2 - 4)}{6x_n^2 - 4} - \frac{2x_n^3 - 4x_n - 3}{6x_n^2 - 4}$$

$$x_{n+1} = \frac{6x_n^3 - 4x_n - 2x_n^3 + 4x_n + 3}{6x_n^2 - 4}$$

$$x_{n+1} = \frac{4x_n^3 + 3}{6x_n^2 - 4}$$

15 (c) Starting with $x_0 = 3$, use the iterative formula given in part (b) to find the value of x_3

Give your answer to **three** decimal places.

[2 marks]

repeatedly sub into formula
3 = ANS

$$\frac{4\text{ANS}^2 + 3}{6\text{ANS}^2 - 4}$$

$$x_0 = 3, x_1 = 2.22, x_2 = 1.82884 \dots$$

$$x_3 = 1.709 \text{ (3dp)}$$

15 (d) Hence find the distance PQ

Give your answer to **two** decimal places.

[2 marks]

$$x_4 = 1.698 \text{ Same to 3dp}$$

$$x_5 = 1.698$$

$x = 1.698$ minimises the distance function.
Sub in $x = 1.698$

$$d^2 = (x^2 - 2.5)^2 + (x - 3)^2$$

$$d^2 = (1.698^2 - 2.5)^2 + (1.698 - 3)^2$$

Turn over for the next question

$$d^2 = 1.8420$$

$$d = \sqrt{1.8420}$$

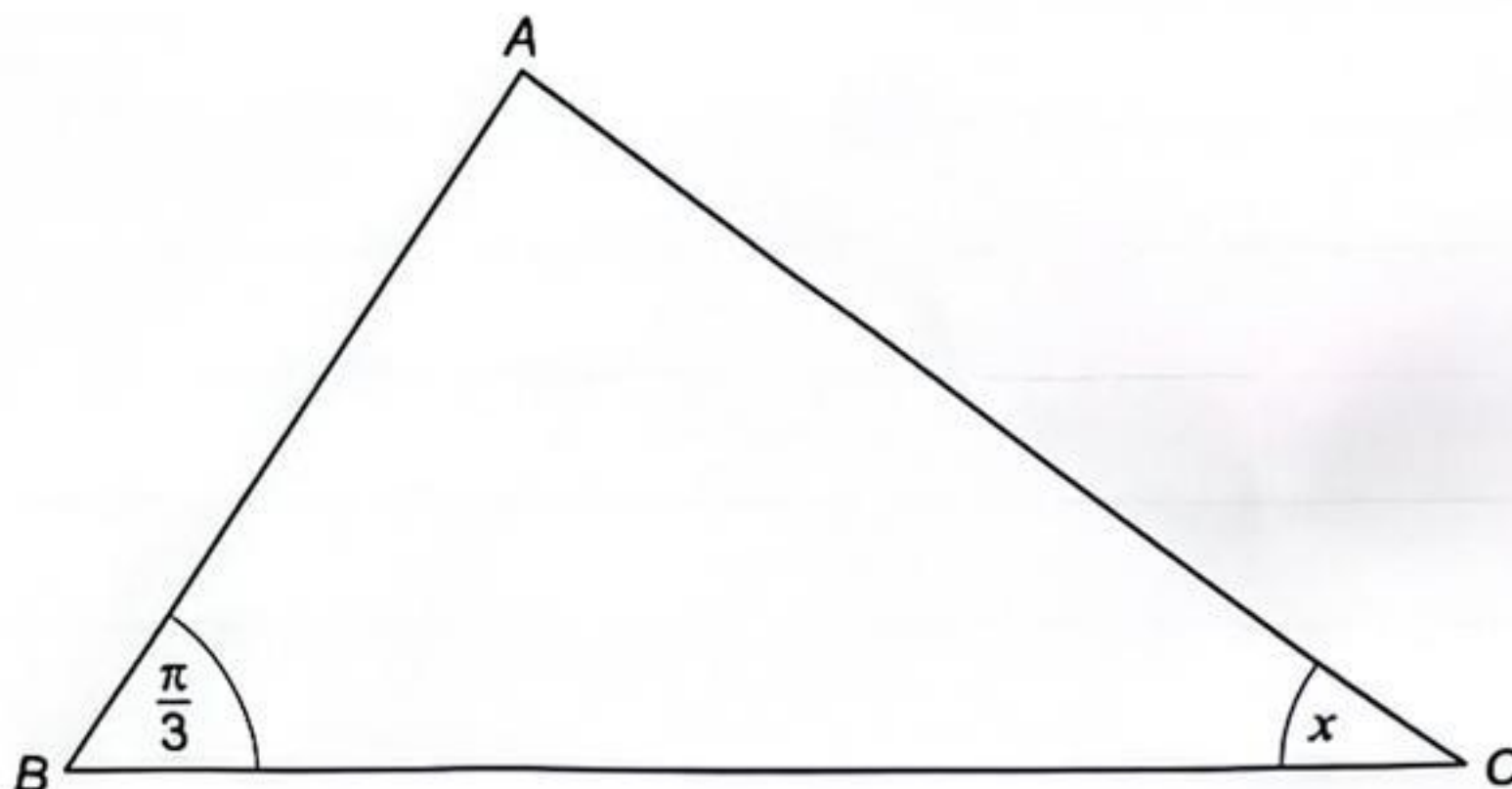
$$d = 1.36 \text{ (2dp)}$$

16

The triangle ABC is shown in the diagram below.

The angle ABC is $\frac{\pi}{3}$ radians.

The angle BCA is x radians.



16 (a)

Explain why angle $BAC = \frac{2\pi}{3} - x$

[1 mark]

total angle in triangle = π

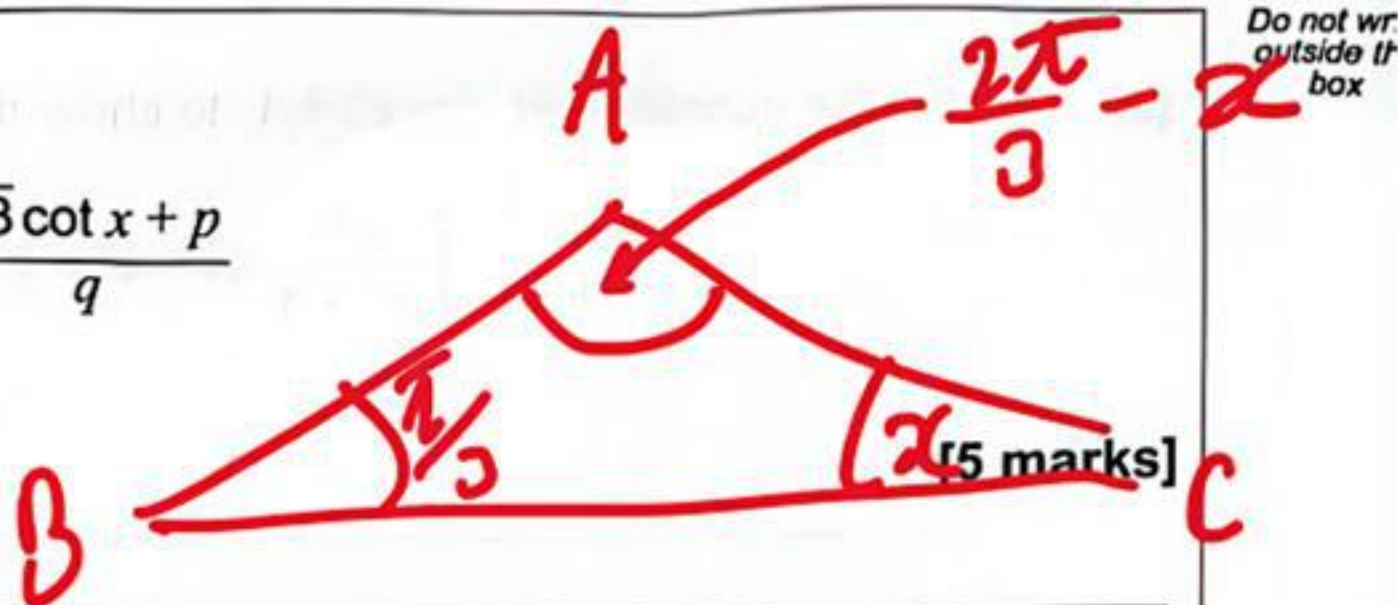
$$BAC = \pi - \left(\frac{\pi}{3} + x\right) = \pi - \frac{\pi}{3} - x$$

$$= \frac{2\pi}{3} - x$$

16 (b) (i) Use the sine rule to show that

$$\frac{BC}{AB} = \frac{\sqrt{3} \cot x + p}{q}$$

where p and q are integers.



$$\frac{BC}{\sin(\frac{2\pi}{3} - x)} = \frac{AB}{\sin(x)} \quad \sin(a-b) = \sin a \cos b - \cos a \sin b$$

$$\frac{BC}{\sin(\frac{2\pi}{3}) \cos x - \cos(\frac{2\pi}{3}) \sin x} = \frac{AB}{\sin x}$$

$$\frac{BC}{\frac{\sqrt{3}}{2} \cos x - -\frac{1}{2} \sin x} = \frac{AB}{\sin x}$$

$$\frac{BC}{AB} = \frac{\frac{\sqrt{3}}{2} \cos x + \frac{1}{2} \sin x}{\sin x} = \frac{\sqrt{3} \cos x + \sin x}{2 \sin x}$$

$$= \frac{\sqrt{3} \cos x}{2 \sin x} + \frac{\sin x}{2 \sin x} = \frac{\sqrt{3}}{2} \cot x + \frac{1}{2} = \frac{\sqrt{3} \cot x + 1}{2}$$

16 (b) (ii) Hence state the exact value of x when

$$\frac{BC}{AB} = \frac{\sqrt{3} + 1}{2}$$

$$\cot x = 1 \quad x = \frac{1}{4}\pi$$

[1 mark]

17 (a)

Use the substitution $u = e^x + 1$ to show that

$$\int \frac{e^{2x}}{e^x + 1} dx = e^x - \ln(e^x + 1) + k$$

[5 marks]

$$\frac{du}{dx} = e^x \quad \frac{dx}{du} = \frac{1}{e^x} \quad dx = \frac{1}{e^x} du$$

$$\int \frac{e^{2x}}{e^x + 1} \times \frac{1}{e^x} du$$

$$= \int \frac{e^x}{e^x + 1} du$$

$$= \int \frac{u-1}{u} du = \int \frac{u}{u} - \frac{1}{u} du$$

$$= \int 1 - \frac{1}{u} du$$

$$= u - \ln u + C$$

$$= e^x + 1 - \ln(e^x + 1) + C$$

$$= e^x - \ln(e^x + 1) + k$$

17 (b)

Solve the differential equation

$$\left(\frac{e^x + 1}{e^{2x}}\right) \frac{dy}{dx} = \cos^2 y$$

Separate variables

given that $y = \pi$ when $x = 0$

Write your answer in the form

$$\tan y = e^x + \ln\left(\frac{A}{e^x + 1}\right) + B$$

$$\frac{1}{\cos^2 y} dy = \frac{e^{2x}}{e^x + 1} dx$$

where A and B are constants to be found.

[5 marks]

$$\int \sec^2 y dy = \int \frac{e^{2x}}{e^x + 1} dx$$

$$\tan y = e^x - \ln(e^x + 1) + k$$

$$y = \pi \text{ when } x = 0$$

$$\tan \pi = e^0 - \ln(e^0 + 1) + k$$

$$0 = 1 - \ln 2 + k \quad k = \ln 2 - 1$$

$$\tan y = e^x - \ln(e^x + 1) + \ln 2 - 1$$

$$\tan y = e^x + \ln\left(\frac{1}{e^x + 1}\right) + \ln 2 - 1$$

$$\tan y = e^x + \ln\left(\frac{2}{e^x + 1}\right) - 1$$

END OF QUESTIONS